



P. A. H. Solapur University, Solapur
First Year B. Tech (All Branches) Semester-I/II

BSC-02 ENGINEERING CHEMISTRY

Teaching Scheme

Theory – 3Hrs. /Week,3Credits

Laboratory – 2 Hrs./Week,1Credit

Examination Scheme

ESE- 70Marks

ISE –30Marks

ICA-25Marks

• **Course Objectives:**

BSC-02.O1	To impart knowledge of importance of quality of water and appropriate water treatment process
BSC-02.O2	To impart knowledge of corrosion & its prevention techniques
BSC-02.O3	To impart knowledge of lubrication and properties of lubrication.
BSC-02.O4	To impart knowledge of quality of fuel and scope of alternate energy sources.
BSC-02.O5	To make students apply knowledge of different polymeric materials.
BSC-02.O6	To equip students with the required analytical skills

• **Course Outcomes:** At the end of this course, students will able to

BSC-02.1	Analyse the water quality and interpret techniques for water purification.
BSC-02.2	Describe lubrication and properties of lubricants, select proper lubricant for specific machine.
BSC-02.3	Explain mechanism of the corrosion, select proper preventive methods of corrosion.
BSC-02.4	Recognize the fuel quality and understand the scope of alternate energy sources.
BSC-02.5	Interpret the chemical synthesis, properties of various polymers and their applications.
BSC-02.6	Describe various types of energy storage systems with their applications.

Section-I

Unit No. 01: Water Chemistry				Hours :08	Marks: 18
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
1.1	Introduction, water quality parameters like pH, acidity, alkalinity, total solids, Dissolved oxygen, chlorides (definitions & permissible limits as per BIS), BOD, COD, (Definition, Determination & Significance).	3	2	Explanation	Understanding remembering
1.2	Hardness: types of hardness (temporary/ permanent), Calcium carbonate equivalent hardness and its calculations. (Numerical problems on hardness),	4	2	Explanation, numerical	Remembering, understanding, applying
1.3	Scale and sludge in boilers: Formation, disadvantages and prevention. Softening of water by Ion exchange process and reverse osmosis process.	6	2	Explanation	Understanding, remembering
1.4	Treatment of water for domestic purpose by aeration, sedimentation with coagulation process. Disinfection of water by chloramine, bleaching powder, chlorine and ozone.	5	2	Explanation, applications	Remembering, understanding
Unit No. 02: Lubricants				Hours: 05	Marks:13
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
2.1	Lubricants & lubrication, functions, classification of lubricants: Solid, semisolid and liquid.	3	1	Explanation	Remembering, understanding
2.2	Characteristic properties of lubricants (only definition) such as viscosity, viscosity index, flash point & fire point, cloud point & pour point, aniline point, oiliness, saponification value & acid value (numerical problems on saponification & acid value)	5	2	Explanation, numerical	Remembering, understanding, applying
2.3	Types of lubrication such as fluid film, boundary (thin film) and extreme pressure. Selection of lubricants for cutting tools, I.C. engine, gears, transformers, delicate instruments & refrigeration system.	5	2	Explanation application	Remembering, understanding

Unit No. 03: Metallic Materials, Corrosion & its Prevention				Hours : 08	Marks:18
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
3.1	<i>Metallic Materials:</i> types of iron: cast iron, steel and wrought iron: composition, properties and applications. <i>Alloys:</i> definition, purposes of making alloys.	4	2	Explanation, application	Remembering, understanding
3.2	<i>Corrosion:</i> definition, classification <i>Dry corrosion:</i> oxidation corrosion: nature of oxide film. <i>Wet corrosion</i> – electrochemical corrosion: Hydrogen evolution mechanism, oxygen absorption mechanism.	5	2	Explanation	Remembering, understanding
3.3	Factors influencing corrosion, testing and measurement of corrosion by weight loss method and electrical resistance method	4	2	Explanation	Remembering, understanding
3.4	<i>Prevention of corrosion by:</i> <i>Cathodic protection & anodic protection</i> <i>Protective coatings:</i> <i>Metal Coating:</i> methods of application of metal coatings such as hot dipping (galvanization & tinning), metal cladding. <i>Organic Coating:</i> Paint: characteristics of good paints, constituents of paints.	5	2	Explanation	Remembering, understanding

Section-II

Unit No. 04: Conventional & Non-conventional Energy Sources				Hours:09	Marks:21
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
4.1	Introduction, classification, characteristics of good fuel, comparison between solid, liquid and gaseous fuel, calorific value (gross and net),	4	2	Explanation,	Remembering, understanding,
4.2	Determination of calorific value by bomb calorimeter and Boy's calorimeter. Dulong's formula for calorific value. (numerical problems on calorific value.)	8	3	Explanation, Application, Numerical	Remembering, understanding, Applying
4.3	Petroleum: introduction, composition, classification, origin, refining of crude oil. <i>Biodiesel:</i> Introduction, preparation, advantages and disadvantages.	5	2	Explanation	Remembering, understanding
4.4	Batteries – Introduction and types of batteries, characteristics of batteries, construction, working and applications of Lithium ion batteries, charging and discharging reactions at respective electrodes.	4	2	Explanation	Remembering, understanding

Unit No. 05: Polymers				Hours :08	Marks:18
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
5.1	Polymerization, types of polymerization, degree of polymerization (DP), numerical problems on degree of polymerization, number average molecular weight (definitions and numerical problems)	6	2	Explanation, application, numerical	Remembering, understanding, applying
5.2	<i>Plastics</i> : definition, properties, types of plastics (thermosoftening and thermosetting), synthesis, properties and applications of PVC and PET, molding of plastic into articles: compression, extrusion and injection & transfer.	6	3	Explanation, application	Remembering, understanding
5.3	<i>Rubber</i> : Classification, isolation of natural rubber, vulcanization, synthesis, properties and applications of Buna-S and Thiokol rubbers. <i>Biodegradable Polymers</i> : Introduction, examples with applications.	6	3	Explanation, application	Remembering, understanding
Unit No. 06: Modern Analytical Techniques				Hours:04	Marks:10
Sr. No.	Subunit	Marks	Hours	Assessment	Bloom's Level
6.1	<i>Concentration of solution</i> :- Normality, molarity, molarity, mole fraction (definition and numerical problems),	4	2	Explanation, application, numerical	Remembering, understanding, applying
6.2	<i>Chromatography</i> : definition, types. <i>GLC</i> : definition, instrumentation and application of GLC. <i>Thermal analysis</i> : definition of TGA, instrumentation and application of TGA.	6	2	Explanation	Remembering, understanding

- **In Semester Evaluation (ISE):**

ISE shall be based upon student's performance in three tests conducted & evaluated at institute level

- **Internal Continuous Assessment (ICA):**

ICA shall be based on minimum **Eight** experiments out of below list

1. Determination of Hardness of water.
2. Determination of Alkalinity of water.
3. Determination of Chloride content in given water.
4. Determination of dissolved oxygen in water.
5. Determination of viscosity of a lubricant by Ostwald's viscometer.
6. Estimation of Copper in Bronze metal alloy.
7. Preparation of phenol formaldehyde resin.
8. Preparation of urea Formaldehyde resin.
9. Determination of COD of water sample.

10. Separation of mixture of organic compounds by TLC.
11. Determination of oil and grease from waste water.
12. Hands on Determination of pH by pH meter.
13. Hands on Determination of EMF of acid base titration by potentiometer.

Books:

1. A text book of Engineering Chemistry, S.S. Dara, S S Umare, S Chand
2. A text book of Engineering Chemistry, Shashi Chawala, Dhanpat Rai & Co
3. A text book of Experiments and Calculations in Engineering Chemistry, S.S. Dara. S Chand

Reference Books:

1. Engineering Chemistry, Jain and Jain, Dhanpat Rai & Co
2. Engineering Chemistry, M. Subha Ramesh, Dr. S. Vairan-Ed.-IIInd Wiley
3. Instrumental Methods of chemical analysis, Chatwal and Anand, Himalaya Pub House
4. Industrial Chemistry, B.K. Sharma, Goyal
5. Chemistry for Engineers, Rajesh Agnihotri, Wiley
6. Fundamentals of Engineering Chemistry, S.K.Singh, New AgeInt.
7. Engineering Chemistry (NPTEL Web book), B. L. Tembe, Kamaluddin & M. S.Krishnan.
8. Fuel Cells-Shripad Revankar, Pradeep Majumdar
9. Fuel Cell Fundamentals-Ryan O'Hayre, Suk-Won Cha, John wiley & Sons
10. Recent trends in Fuel Cell Science and Technology-Suddhasatwa Basu, Anamaya Publishers, New Delhi.